MINING TECHNOLOGY **Course Code : 332305**

Programme Name/s	**: Mining & Mine Surveying/ Mining Engineering**
Programme Code	**: MS/ MZ**
Year	**: Second**
Course Title	**: MINING TECHNOLOGY**
Course Code	**: 332305**

I. RATIONALE

The students of Mining and Mine surveying must know basic mining operations carried out in underground Coal and Metal Mines. The knowledge of Drilling, Charging, Explosives and detonators used for underground Blasting is essential for Mining Diploma Holders. As a Overmen /Sirdar diploma holder is in charge of the under ground mining district and responsible for safety of persons working under him . The knowledge of different types of roof supports will help mining engineer to choose appropriate method of supporting to make working safe. The course will also provide useful information to students for shot firing and ground control in underground coal and metal mines.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences: Select relevant type of support for different ground conditions and undertake the blasting operation at face as per regulation.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Choose relevant explosives and accessories used for underground blasting.
- CO2 - Prepare the face for solid blasting in underground mines.
- CO3 - Carry out the process of shot firing safely in underground mines.

- CO4 - Install and withdraw the different types of conventional supports used in underground mines.
- CO5 - Install different types of roof bolt and cable bolts in underground mines.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
332305	MINING TECHNOLOGY	MTE	DSC	2	-	2	-	4	4	1.5	30	70*#	100	40	25	10	25#	10	-	-	150

Total IKS Hrs for Sem. : 1 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.

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1	TLO 1.1 Classify given types of explosives based on their properties TLO 1.2 Select suitable type of Permitted explosives in given situation TLO 1.3 Select suitable type of detonators in a given situation. TLO 1.4 Choose suitable type of safety fuse, detonating cords, detonating relays in given situation TLO 1.5 Suggest proper types of shot firing tools and exploders for particular application.	**Unit - I Explosives And Blasting Accessories** 1.1 Ancient explosive and blasting process (IKS), Common explosive bases, Properties of Explosives, High Explosive & Low explosive, and their comparison. 1.2 Permitted explosives their types, composition, properties, uses, advantages & disadvantages. Brand names of some commonly used explosive of each type. 1.3 A detonator, common types of detonators, plain detonators, instantaneous and delay action detonators their construction, uses, comparison etc. low tension & high-tension detonators. 1.4 Safety fuses, detonating cords, detonating relays. 1.5 Applications of various Shot firing tools, and exploders	Presentations Lecture Using Chalk-Board Lecture Using Chalk-Board Video Demonstrations Model Demonstration

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Interpret given drilling patterns for shot firing. TLO 2.2 Explain proper sequence of shot firing procedure. TLO 2.3 Follow the precautions while blasting-off-solid. TLO 2.4 Describe Historical Blasting Practices.	**Unit - II Blasting Practices in U/G Mines** 2.1 Drilling patterns for shot firing on machine cut face, in stone drift etc. 2.2 Face preparation for shot firing, Preparation of priming charge, charging of hole in coal and rock in underground working only, Direct and inverse initiation, shot firing circuits, procedure of shot firing of holes in gassy mine, precautions. Simultaneous & delay firing. 2.3 Solid blasting, conditions to be satisfied before doing solid blasting, advantages of solid blasting, drilling patterns used with solid blasting 2.4 The use of explosives in mining in ancient time and the standard method for blasting rocks was to drill a hole to a considerable depth and deposit a charge of gunpowder.	Video Demonstrations Video Demonstrations Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Calculate detonator factor and powder factor. TLO 3.2 Identify causes of Misfire shot. TLO 3.3 Identify stemming materials and their purpose. TLO 3.4 Interpret the explosive magazines. TLO 3.5 Describe the procedure of disposal of out dated explosives.	**Unit - III Safety in Blasting** 3.1 Charge of explosive required for blasting in coal, rock. Powder factor, detonator factor. Precaution to improve blasting results. 3.2 Misfires, causes, remedy and method of relieving dealing with misfires, blown out shots, blown through shots causes and precautions. 3.3 Purpose of stemming, Stemming materials used for shot firing, water ampoules for stemming. 3.4 Storage of explosives, Magazines 3.5 Disposal of outdated explosives.	Lecture Using Chalk-Board Video Demonstrations Presentations Video Demonstrations Video Demonstrations
4	TLO 4.1 Explain various types of support used in Mines. TLO 4.2 Identify given types of wooden support Prop, chock, crossbar, fore poling, safari support. TLO 4.3 Describe the installation and Withdrawal of support. TLO 4.4 Interpret support density as per given support plan.	**Unit - IV Conventional Roof Support** 4.1 Classification of different types of supports used in Underground Mines. 4.2 Different types of Wooden Supports used in Underground Mines (Description of Prop , chock, crossbar, fore poling , safari support, only) 4.3 Installation and Withdrawal of support. 4.4 Support Plan and support density.	Lecture Using Chalk-Board Video Demonstrations Video Demonstrations Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Apply the theory of mechanics of strata behavior for stability of underground opening. TLO 5.2 Describe the use of various types of roof bolt supports based on their principle of action TLO 5.3 Follow the specific code of practice for roof bolting. TLO 5.4 Describe the procedure of roof stitching. TLO 5.5 Describe the procedure of cable Bolting TLO 5.6 Classify given types of yielding props. TLO 5.7 Identifygiven types of friction props and hydraulic props.	**Unit - V Roof Bolting, Cable Bolting and Yielding Prop** 5.1 Theories of mechanics of strata behavior: Dome or arch theory, Beam theory. 5.2 Roof Bolting: Theories, Principle of action, functions, Types (Cement capsule grouted bolts, Resin bolt). 5.3 Anchorage testing of Roof Bolts, Code of practice for roof bolting in underground mines. 5.4 Roof stitching, Principle of Roof stitching. 5.5 Cable Bolting, W-strap. 5.6 Yielding supports, Classification .important definition like setting load, 5.7 Different types of friction prop and Hydraulic Props and their installation and withdrawal.	Presentations Video Demonstrations Presentations Video Demonstrations Video Demonstrations Lecture Using Chalk-Board Model Demonstration

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify different Types of permitted Explosives Cartridges.	1	*Identifcation of permitted Explosive Cartridges. (P1-Permitted Explosive,P2-Sheathed Explosive,P3-Equivalent Sheathed Explosive,P4-Extra Safe Explosive,P5-Explosive of Solid Blasting.	2	CO1
LLO 2.1 Use the cut section of Instantaneous Electric Detonator	2	*Demonstration of cut section of Instantaneous Electric Detonator. (Neoprene Plug,Flashing Mixture,Prime Charge (ASA),Base Charge (PETN).	2	CO1
LLO 3.1 Use the cut section of delay Detonator.	3	*Demonstration of cut section of delay Detonator. . (Neoprene Plug,Flashing Mixture,Delay Element,Prime Charge (ASA),Base Charge (PETN).	2	CO1
LLO 4.1 Use the different types of shot Firing tools.	4	*Identification of different shot Firing tools.(Electric Lamp/Torch, Stop Watch, Flame Safety Lamp, Wooden Stemming Rod, Scraper, Crack Detector, Wooden Pricker , Sharp Knife and Crimper).	2	CO2
LLO 5.1 Use different drilling patterns for shot firing.	5	*Demonstration of different drilling patterns for shot firing. (Concentric Rings, Pyramid Cut, Diamond Cut, Wedge Cut, Fan Cut, Drag Cut, Burn Cut and Coromant Cut.)	2	CO2
LLO 6.1 To Prepare and Understand Prime Cartridge.	6	*Prepare primer cartridge for solid blasting. (Using Explosive Cartridge and Detonator).	2	CO2
LLO 7.1 Identify the direct initiation and indirect initiation of blast hole.	7	*Demonstration of direct initiation and indirect initiation of blast hole (Direct Initiation for coal mines and Indirect Initiation for drivages of shaft, drifts and tunnels).	2	CO2
LLO 8.1 Identify series and parallel connection for blast holes.	8	*Preparation of Blasting Circuits (Series and Parallel).	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 9.1 Calculate detonator factor and powder factor under given conditions/ problems.	9	*To determine detonator factor and powder factor under given conditions/ problems.	2	CO3
LLO 10.1 Identify Misfire shot and carry out the procedure to deal Misfire shot.	10	*Identification of Misfire shot and carry out the procedure to deal Misfire shot.	2	CO3
LLO 11.1 Draw a layout of Magazine under given conditions.	11	*Preperation of layout of Magazine under given conditions.	2	CO3
LLO 12.1 Install the prop support (with the help of model) under given condition.	12	*Installation of Prop Support.	2	CO4
LLO 13.1 Install the chock/cog support (with the help of model) under given condition.	13	*Installation of Chock/Cog Support.	2	CO4
LLO 14.1 Install the crossbar, and safari support under given condition.	14	*Installation of Cross bar and Safari Support.	2	CO4
LLO 15.1 Carryout the appropriate steps for fore poling method of Support.	15	*Installation of Fore Poling Method of Support	2	CO4
LLO 16.1 Withdraw the support using Sylvester prop withdrawer with the help of model.	16	*Demonstrate the procedure of withdrawal of support.	2	CO4
LLO 17.1 Draw support plan and frame SSR under given conditions.	17	*Preparation of support plan and apply of SSR	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 18.1 Identify different types of roof bolts	18	*Demonstration of Roof Bolts	2	CO4 CO5
LLO 19.1 Develop the procedure to install cement capsule (quick setting cement) roof bolts using video clip and virtual media.	19	Installation of Cement Capsule Roof Bolt	2	CO5
LLO 20.1 Develop the procedure to install resin capsule roof bolts using video clip and virtual media.	20	Installation of Resin Capsule Roof Bolts	2	CO5
LLO 21.1 Develop the procedure to install Roof stitching using video clip and virtual media.	21	Installation of Roof Stitching	2	CO5
LLO 22.1 Develop the procedure to install Cable bolting using video clip and virtual media.	22	Installation of Cable Bolting	2	CO5
LLO 23.1 Install Friction Prop (with the help of model) under given condition.	23	Installation of Friction Prop	2	CO5
LLO 24.1 Install hydraulic prop (with the help of model) under given condition.	24	Installation of Hydraulic Prop	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- •Detonators: Prepare report of various Detonators used in that mine by visiting any underground coal mine.
- Prepare a report on Safety Fuse, Detonating cords and Detonating Relays used in that mine by visiting any underground coal mine.
- Explosives and Blasting Accessories: Prepare report of various explosives and accessories used in that mine by visiting any underground coal mine.
- Prepare a Report on Roof Bolting and Cable Bolting.
- SSR : Collect the copy of Systematic Support Rules of nearby mines and calculate support density.
- Conventional Roof Support : Prepare the cut out Model of Different types of support used in mines.
- Safety in Blasting: Collect actual date for measuring pull, Detonator factor and powder Factor.
- Prepare a report on Prime cartridge ,Direct Initiation and Inverse Initiation .
- Blasting Patterns: Prepare a Report on Various types of Blasting /Drilling Patterns.
- Prepare a Report on various shot firing tools and Exploders used in that mine by visiting any underground coal mine.
- Yielding Support : Collect data on different types of Yielding Support used in Indian Mines ,their Load bearing capacity and manufacturers.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Dummy cartridges of different Types of permitted Explosives Cartridges.	1,6,8
2	Flame safety lamp with dark room facility.	10
3	Models of the prop, chock, crossbar, and safari support.	12,13,14
4	Model of fore poling method of Support	15
5	Sylvester prop withdrawer.	16
6	Model of cement capsule (quick setting cement) used in roof bolts.	18,19
7	Cut out models of Instantaneous Electric Detonator	2
8	Model of different types resin capsule used in roof bolts.	20
9	Model of Friction Prop	23
10	Model of Hydraulic Prop	24

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
11	Cut out models of delay Detonator used for Shot firing in underground Mines.	3
12	Different shot Firing tools.	4
13	Crack detector used for detecting cracks in shot holes.	4,5,7
14	Templates for different drilling patterns used for shot firing.	5
15	Single Shot and multi shot exploder.	7,8,10
16	Smart Classroom fitted with computer and DLP	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Explosives And Blasting Accessories	CO1	12	2	6	6	14
2	II	Blasting Practices in U/G Mines	CO2	12	2	6	6	14
3	III	Safety in Blasting	CO3	10	4	4	6	14
4	IV	Conventional Roof Support	CO4	10	2	4	4	10
5	V	Roof Bolting, Cable Bolting and Yielding Prop	CO5	16	4	6	8	18
Grand Total				**60**	**14**	**26**	**30**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Assignment,Self-Learning and Team Work
- Mid Term Test
- Rubrics for COs

- Seminar/Presentation

Summative Assessment (Assessment of Learning)

- Lab Performance
- End Of Term Examination
- Viva-voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	1	2	3	1	3	3			
CO2	3	2	2	3	1	3	3			
CO3	3	3	2	3	1	3	3			
CO4	3	1	3	3	1	3	3			
CO5	3	1	2	3	1	2	3			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	G.K. Pradhan	Explosive and Blasting Technology	Lovely Prakashan, Dhanbad,ISBN-978-9058096050
2	L.C.Kaku	Mining Digest	Lovely Prakashan, Dhanbad,ISBN Number-13. 978-8179561515
3	Dr.B.P.Verma	Roof Bolting	Lovely Prakashan, Dhanbad,ISBN-13. 978-8179561652
4	D.J.Deshmukh	Elements of Mining Technology	Lovely Prakashan, Dhanbad,ISBN-818990433-7
5	R.D.Singh	Principles & Practices of Modern Coal Mining	New Age International (P) Limited Publishers,New Delhi,ISBN Number-81-224-0974-1
6	Heartman L.Howard	Introductory Mining Engineering	John Wiley & Sons,New York,ISBN Number-0471-62804-2

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.slideshare.net/search?searchfrom=header&q=explosives	Regarding Mining
2	https://www.youtube.com/results?search_query=mining+technology	For Videos on mining
3	https://drive.google.com/file/d/1c-LHOVpvEoySm_68wbGeyVoEHHtUcl6u/view	Explosive and blasting Related Information
4	https://www.e-education.psu.edu/geog000/node/827	Regarding mining videos and books

Sr.No	Link / Portal	Description
5	https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=8zYwEsyFCoiPyJlPmzHDxg==	Mining Related Information

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students